Individual Assignment 5

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1. Set up

Packages

```
# general use and cleaning
library(tidyverse)
library(here)
library(janitor)
library(snakecase)

# wrapping axis labels
library(scales)

# reading in .xlsx files
library(readxl)

# visualizing missing data
library(naniar)

# arranging plots
library(patchwork)
```

Data

Question 1

A: Planning

- x axis: Site

- y axis: log transformed Ostracod abundance
- geometry abundance: geom_jitter()
- geometry median: stat_summary()

B: Wrangle Data

```
#new object from aq_ins_clean
ostracod <- aq_ins_clean |>
  # filter to only include ostracod
  filter(taxon == "ostracod") |>
  # log + 1 transform mean abundance per liter
  mutate(log_ave_lit = log(ave_lit + 1))

# display 10 random rows
slice_sample(ostracod, n=10)
```

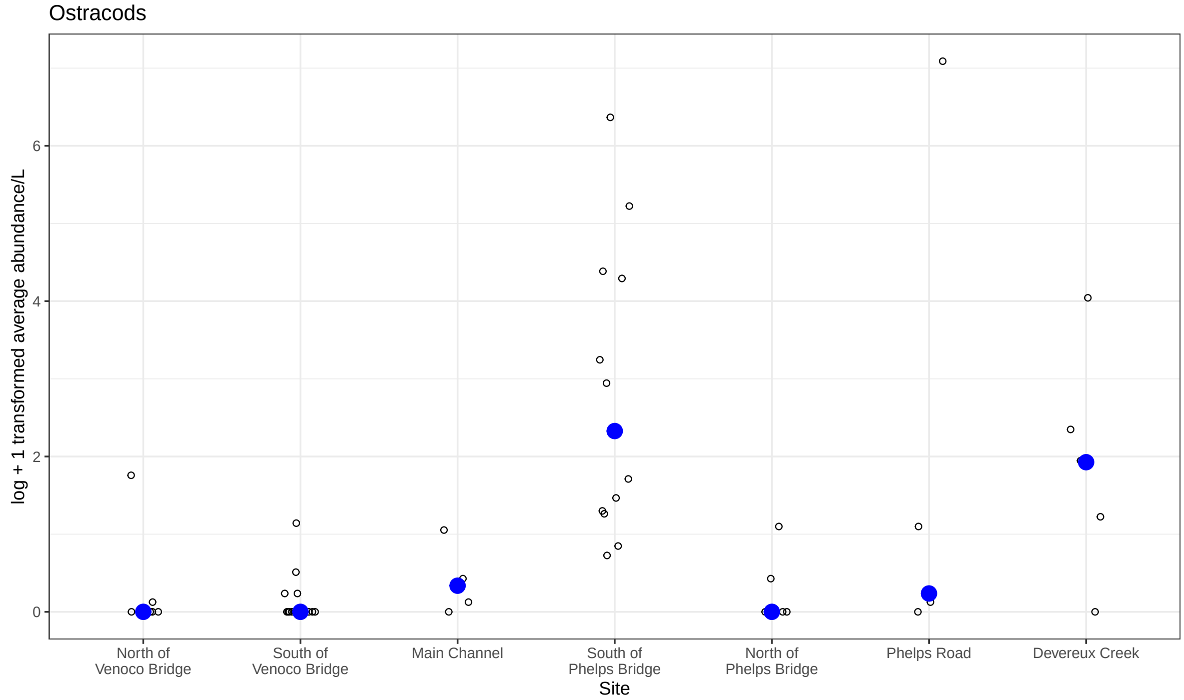
```
# A tibble: 10 x 24
   date_site       date                site  taxon ave_lit med_ph med_sal med_do
   <chr>           <dttm>              <chr> <chr>   <dbl>  <dbl>   <dbl>  <dbl>
 1 2022-05-12_NPB  2022-05-12 00:00:00 NPB   ostr~   1.07    7.97   11.4    8.17
 2 2023-08-12_NEC  2023-08-12 00:00:00 NEC   ostr~   0      NA      NA     NA
 3 2023-03-03_NPB2 2023-03-03 00:00:00 NPB2  ostr~   0.133   7.71    1.98   9.59
 4 2022-05-13_NVBR 2022-05-13 00:00:00 NVBR  ostr~   0       9.72   74.2   11.4
 5 2021-11-12_NPB  2021-11-12 00:00:00 NPB   ostr~  79.2    NA      NA     NA
 6 2020-02-21_NVBR 2020-02-21 00:00:00 NVBR  ostr~   4.8     7.34    4.1    6.1
 7 2020-01-21_NDC  2020-01-21 00:00:00 NDC   ostr~   2.4     6.53    0.3    7.6
 8 2022-01-21_NEC  2022-01-21 00:00:00 NEC   ostr~   0      NA      NA     NA
 9 2022-11-18_NVBR 2022-11-18 00:00:00 NVBR  ostr~   0      NA      NA     NA
10 2022-08-12_NEC  2022-08-12 00:00:00 NEC   ostr~   0       7.37   NA     NA
# i 16 more variables: taxonRank <chr>, scientificName <chr>, kingdom <chr>,
#   Phylum <chr>, Class <chr>, Subclass <chr>, Superorder <chr>, Order <chr>,
#   Infraorder <chr>, Family <chr>, Subfamily <chr>, Tribe <chr>, Genus <chr>,
#   comments <chr>, site_full <fct>, log_ave_lit <dbl>
```

C: Code Figure

```
# base layer
ostracods_plot <- ggplot(data = ostracod,
```

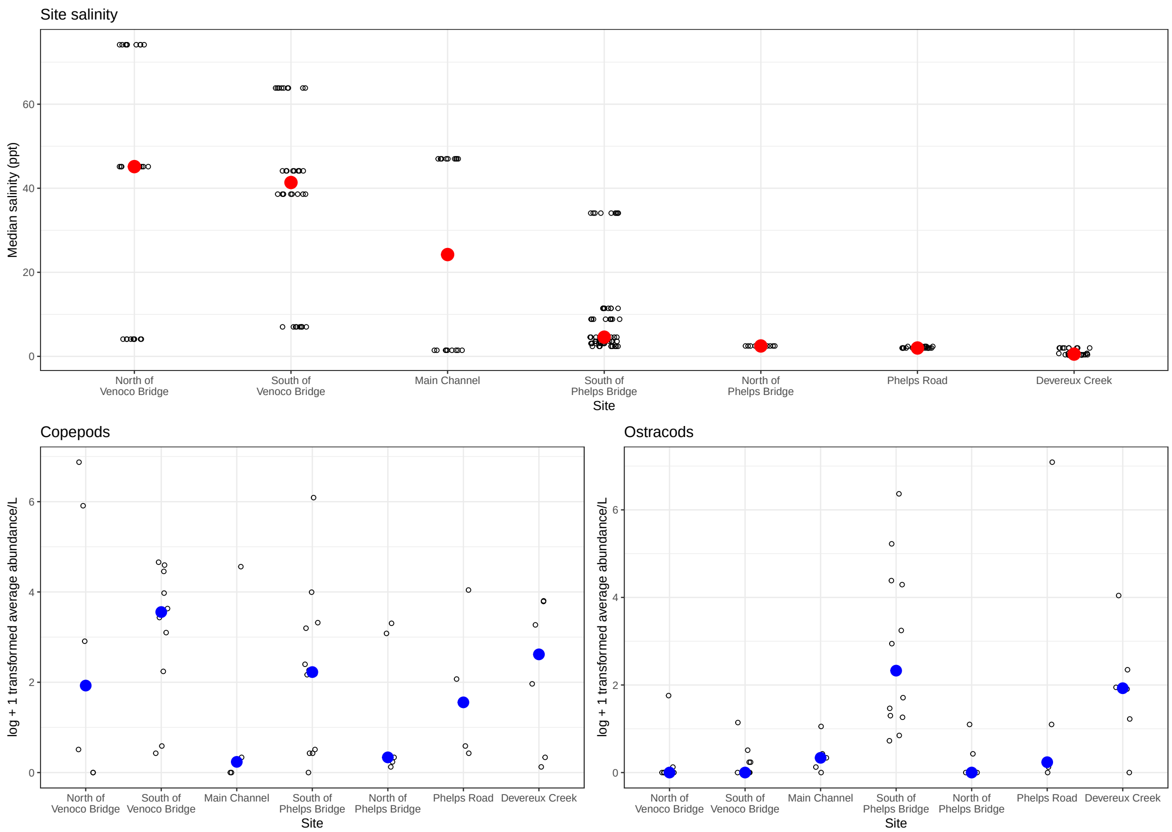
```r
                   # x-axis: site
                   # y-axis: mean abundance/L
                   mapping = aes(x = site_full,
                                 y = log_ave_lit)) +
  # first layer: points to represent observations
  geom_jitter(height = 0,
              shape = 21,
              width = 0.1) +
  # second layer: points to represent medians
  stat_summary(geom = "point",
               fun = "median",
               size = 4,
               color = "blue") +
  # wrapping x-axis text
  scale_x_discrete(labels = label_wrap(15)) +
  # relabelling x- and y-axis, adding title
  labs(x = "Site",
       y = "log + 1 transformed average abundance/L",
       title = "Ostracods") +
  # different theme
  theme_bw()

# displaying object
ostracods_plot
```



D: Create multipanel figure

```
#combining plots
salinity/(copepods_plot+ostracods_plot)
```



E: Analysis

- The differences in sites between copepods and ostracod abundance are mainly at the South of Phelps Bridge as ostracods are the most abundant here while copepods are more abundant South of Venoco Bridge.
- Ostracod abundance may not be related to site-level differences as there are numerous occassions of median salinity being relatively low at places like Phelps Bridge or Phelps road, yet South of Phelps Bridge sites have high abundance while the North side doesn't. Additionally,low abundance sites like North of Phelps have the same abundance as high salinity sites like Venoco Bridge.

Question 2

A: Planning

- x axis: salinity

- y axis: log transformed abundance
- color: taxon
- geometry: geom_point()
- facets: facet_Wrap(~taxon, scales="free")

B: Wrangle Data

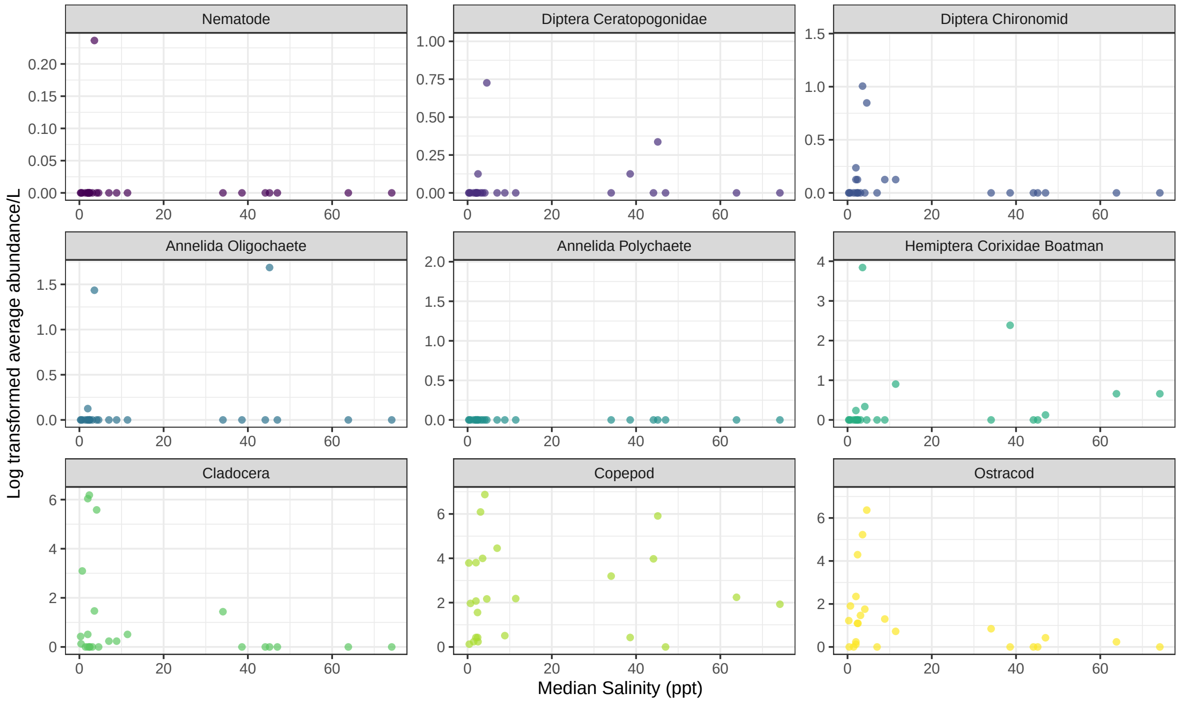
```
# creating new clean object from aquatic inverts
ab_salinity <- aq_ins_clean |>
  # log transform mean abundance per liter
  mutate(log_ave_lit = log(ave_lit+1),
         # convert taxon to title
         taxon = to_any_case(taxon, case = "title"),
         # order taxon levels by abudance
         taxon = fct_reorder(taxon, ave_lit, .fun="max"))

#display 10 random rows
slice_sample(ab_salinity, n = 10)
```

```
# A tibble: 10 x 24
   date_site       date                site  taxon ave_lit med_ph med_sal med_do
   <chr>           <dttm>              <chr> <fct>   <dbl>  <dbl>   <dbl>  <dbl>
 1 2022-01-21_NEC  2022-01-21 00:00:00 NEC   Cope~    98    NA      NA     NA
 2 2023-02-12_NEC  2023-02-12 00:00:00 NEC   Anne~     0    NA      NA     NA
 3 2022-11-19_NPB1 2022-11-19 00:00:00 NPB1  Nema~     0     7.26    2.47   3.96
 4 2023-05-04_NEC  2023-05-04 00:00:00 NEC   Dipt~     0     8.05   NA      8.55
 5 2022-05-12_NPB  2022-05-12 00:00:00 NPB   Dipt~     0     7.97   11.4    8.17
 6 2023-05-16_NVBR 2023-05-16 00:00:00 NVBR  Anne~     0     9.29   NA      9.83
 7 2022-01-21_NEC  2022-01-21 00:00:00 NEC   Anne~     0    NA      NA     NA
 8 2023-07-17_NPB  2023-07-17 00:00:00 NPB   Dipt~     3.2  NA      NA     NA
 9 2023-07-17_NPB  2023-07-17 00:00:00 NPB   Clad~     0    NA      NA     NA
10 2022-11-19_NPB1 2022-11-19 00:00:00 NPB1  Anne~     0     7.26    2.47   3.96
# i 16 more variables: taxonRank <chr>, scientificName <chr>, kingdom <chr>,
#   Phylum <chr>, Class <chr>, Subclass <chr>, Superorder <chr>, Order <chr>,
#   Infraorder <chr>, Family <chr>, Subfamily <chr>, Tribe <chr>, Genus <chr>,
#   comments <chr>, site_full <fct>, log_ave_lit <dbl>
```

C: Code figure

```r
# base layer
ggplot(data = ab_salinity,
       # x-axis: site
       # # y-axis: mean abundance/L
                   mapping = aes(x = med_sal,
                                 y = log_ave_lit,
                                 color = taxon)) +
  # layer 1: points to represent observations
  geom_point(alpha = 0.7) +
  #facet wrap for taxon
  facet_wrap(~ taxon,
             scales = "free") +
  # use color palette
  scale_color_viridis_d() +
  # labels
  labs( x = "Median Salinity (ppt)",
        y = "Log transformed average abundance/L") +
  # theme
  theme_bw()+
  # remove legend from theme
  theme(legend.position = "none")
```



D: Analysis

- The differences between taxa in relationships between abundance and salinity is that the higher the salinity, the lower the average abundance. However, this isn't the case with all species as Copepods are abundant at high and low salinities, similar to Hemiptera as both show up to 2.5 log average abundance/L in higher salinities. On the other hand, species like Annelida Oligochaete, along with the majority of the species, follow an exponential decrease as salinity increases. This includes Cladocera and Diptera Chrionomid to name a few.